

Exercise 4: Moments and Estimators

*Teaching Assistant: Sabri Meyer**Lecturer: Foivos Alimisis***Problem 1 (Independence of Exponentials):**

Prove that if X and Y are independent random variables, then e^{tX} and e^{tY} are also independent for any constant $t \in \mathbb{R}$.

Problem 2 (Moments of the Normal Distribution):

Compute the first four moments of the normal distribution $N(\mu, \sigma^2)$.

Problem 3 (Equality in Jensen's inequality):

Let X be a random variable with finite mean $\mathbb{E}[X] = \mu$. Let ϕ be a strictly convex function (i.e. $\phi''(x) > 0$). Prove that the equality in Jensen's inequality, $\mathbb{E}[\phi(X)] = \phi(\mathbb{E}[X])$, holds if and only if X is a constant with probability 1.

Problem 4 (MLEs of Geometric and Exponential Distributions):

Compute the maximum likelihood estimators of the parameters p (probability of success) of the geometric distribution and λ (rate of events) of the exponential distribution. Study whether these estimators are biased.

Problem 5 (MLE for Poisson distribution):

Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with parameter λ . The Poisson distribution is a discrete distribution that models the number of events in a fixed interval that happen with a certain average rate. The PMF is: $\mathbb{P}(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$. Find the MLE for λ and the MLE for $\psi = \mathbb{P}(X = 0)$. Is the estimator for ψ unbiased?

Problem 6 (Estimators for a shifted exponential distribution):

Consider a random sample $X_1, \dots, X_n$ from a shifted exponential distribution with the PDF:

$$f(x; \lambda, \mu) = \lambda e^{-\lambda(x-\mu)} \quad \text{for } x \geq \mu$$

and 0 otherwise. Find the MOM and MLE estimators for λ and μ .